Fuzzy Sets , Logic and System & Applications

Week 2 : Assignment 2 Solution

Q.1. Which of the following is a disadvantage of using Open Right membership functions?

(a) Lack of expressiveness

(b) Difficulty in implementation

(c) Limited applicability

(d) Potential for uncertainty on the right side

(e) None of the above

Solution: (d)

Q.2. What does an Open Left membership function imply about the leftmost part of the function?

(a) It has a sharp peak

(b) It has a gradual slope

(c) It extends infinitely to the left

(d) None of the above

Solution: (c)

Q.3. What characterizes an Open Left membership function?

(a) It has a sharp peak on the left side

(b) It extends infinitely to the left

(c) It has a gradual slope on the left side

(d) None of the above

Solution: (b)

Q.4. An Open Right membership function is characterized by:

(a) An infinite extension to the right

(b) An abrupt cut-off on the right side

(c) Being symmetric around the peak

(d) None of the above

Solution: (a)

Q.5. Given an Open Left membership function with a peak at x = 4, if the degree of membership at x = 2 is 1.0, what is the degree of membership at x = 1?

(a) 0.2

(b) 0.4

(c) 1.0

(d) None of the above

Solution: (c)

Q.6. In a Left-Right membership function defined by parameters [a=2, b=3, c=5], what does parameter 'c' represent?

(a) The width of the left region

(b) The width of the right region

(c) The point where the membership value is 1

(d) None of the above

Solution: (c)

Q.7. If you have two Left-Right membership functions with parameters [1, 4, 6] and [3, 2, 7], which function has a wider right region?

(a) [1, 4, 6]

(b) [3, 2, 7]

(c) Both have the same right region width

(d) None of the above

Solution: (b)

Q.8. In a Gaussian membership function, what does the parameter 'μ' represent?

(a) The standard deviation

(b) The mean

(c) The maximum membership values

(d) The overall width of the function

(e) None of the above

Solution: (b)

Q.9. For two Gaussian membership functions with parameters $\mu_1=3$, $\sigma_1=2$ and $\mu_2=5$, $\sigma_2=5$, which statement is true?

(a) The width of the first function is greater

(b) The width of the second function is greater

(c) Both functions have the same width

(d) It cannot be determined from the given information

(e) None of the above

Solution: (b)

Q.10. If the standard deviation (σ) of a Gaussian membership function is increased, what impact does it have on the function?

(a) The peak becomes sharper

(b) The peak becomes broader

(c) The mean shifts to the left

(d) The mean shifts to the right

(e) None of the above

Solution: (b)